

An exact certificate excluding fifteen palindrome blocks on five labels

Research computation, 30 September 2026

Proposition 1. *No concatenation of fifteen blocks of the form $\rho \operatorname{rev}(\rho)$, where each ρ is a permutation of five labels, is permutation-fair.*

This is a restricted construction obstruction. It does not rule out arbitrary five30-sided permutation-fair dice. Its literature priority has not been established: earlier work already mentions exclusions of structured subclasses.

Proof by an exact integer certificate. Work in the associative algebra in which words with a repeated letter vanish. Write $S = X_0 + \cdots + X_4$ and let P_ρ be the subsequence signature of the palindrome block. Every block is two-wise fair, so

$$P_\rho = \exp(2S) + D_\rho, \quad \deg D_\rho \geq 3.$$

Products of two defects have degree at least six. Thus, for blocks at positions $i = 0, \dots, 14$, the degree-five defect of their product is exactly the sum of

$$\exp(2iS)D_\rho \exp(2(14-i)S).$$

For a distinct-letter pattern q , let $N_\rho(q)$ be its subsequence count in the block and put

$$d_3(q) = 3N_\rho(q) - 4, \quad d_4(q) = 3N_\rho(q) - 2, \quad d_5(q) = 15N_\rho(q) - 4.$$

For each length-five pattern $p = p_1 \cdots p_5$ and $j = 14 - i$, the integer column recording fifteen times this transported defect is

$$\begin{aligned} c_{i,\rho}(p) = & d_5(p) + 10i d_4(p_2 \cdots p_5) + 10j d_4(p_1 \cdots p_4) \\ & + 10i^2 d_3(p_3 p_4 p_5) + 20ij d_3(p_2 p_3 p_4) + 10j^2 d_3(p_1 p_2 p_3). \end{aligned}$$

Fairness therefore requires $\sum_i c_{i,\rho_i} = 0$. Global relabeling allows us to fix $\rho_0 = (0, 1, 2, 3, 4)$.

The accompanying file `five_30_farkas.json` supplies an integer vector λ indexed by the120 patterns, with all entries between -20 and 20 . Direct integer evaluation gives

$$\lambda \cdot c_{0,\text{id}} + \sum_{i=1}^{14} \min_{\rho \in S_5} \lambda \cdot c_{i,\rho} = 40080 > 0.$$

Every proposed sequence consequently has a strictly positive scalar product of its total defect with λ , and cannot have zero total defect. The independent program `verify_five_30_obstruction.py` regenerates all1680 remaining columns and verifies the displayed integer inequality. It needs only the Python standard library, no numerical solver. $\square$

Additional verification. An independent integer subsequence dynamic program checked the transported-defect formula on20 random150-letter words and all120 full patterns in each. The linear-programming search was used only to find the certificate; correctness rests on the subsequent integer checks.